

Worksheet 8

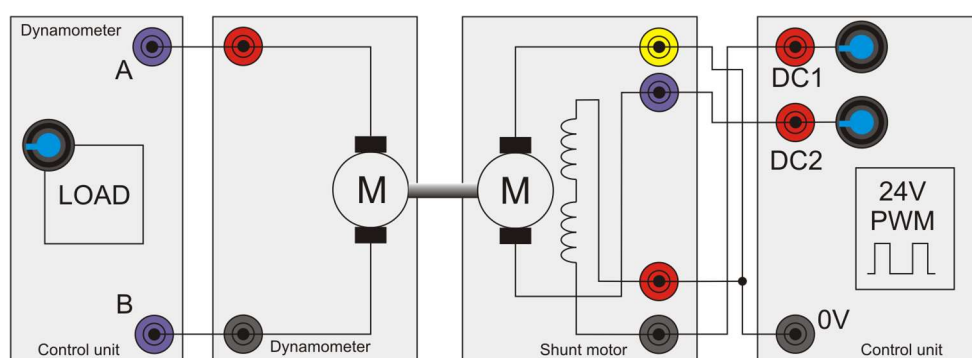
Shunt-wound motor characteristics

Modern electrical machines



Speed control is an important element of electromechanical system design.

Some motors are better than others at self-regulating speed. The shunt motor is really good at speed regulation and for this reason they are often used in devices like lathes, which need to run at a constant speed.



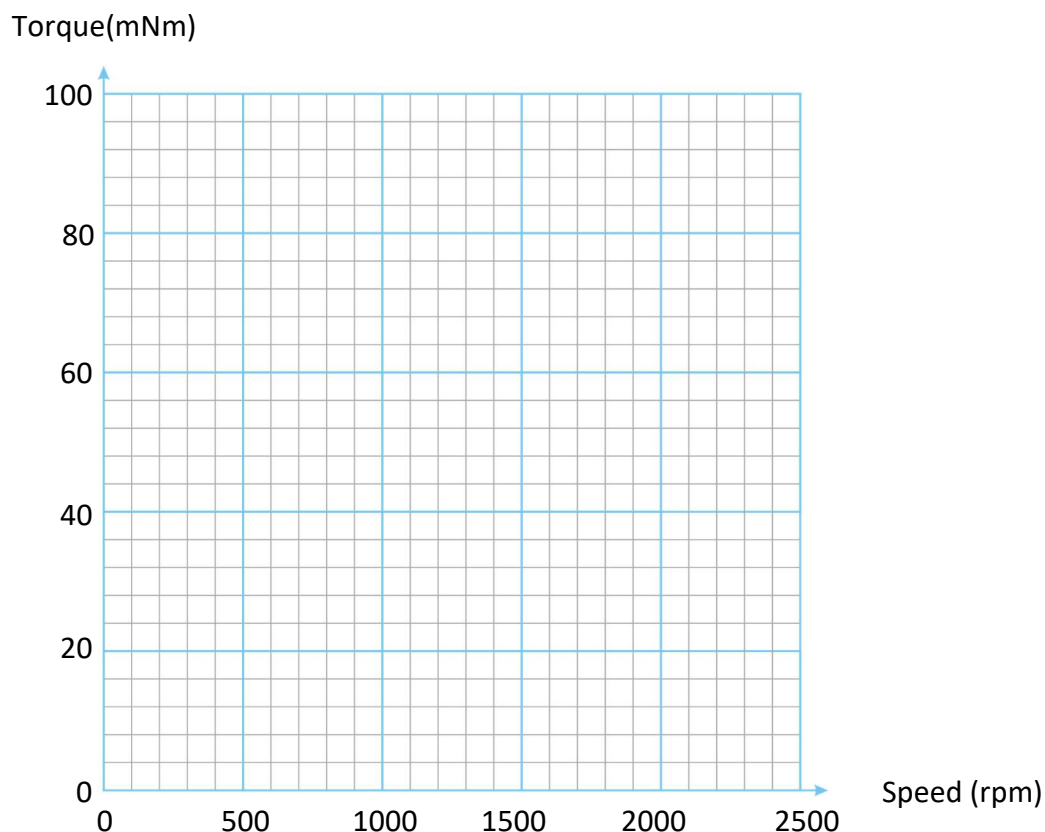
Over to you:

- 1) Set up the shunt motor as shown above.
- 2) On your PC, run the program 'Open Log_DC.bat'.
- 3) Open the program. Set the DC1 output to 100% to power the **armature** and the DC2 output to around 50% for the **field** winding .
- 4) Run the program and press 'Sweep' to take an automatic set of readings for a range of dynamometer loads.
- 5) Repeat steps 4 and 5 for **field** winding DC outputs of 60%, and 70%.
- 6) When the sequence has finished load the results into Excel and use Excel to plot a graph of speed and torque.
- 7) On the next page, use the axes provided to sketch the results.

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The effect of speed on torque for different values of field coil voltage.

Over to you:

- 1) With the same equipment set up, load the program 'Open Control_DC.bat'
- 2) Route the Dynamometer output to the control unit Voltmeter. The software gives Dynamometer current. Now you can get Dynamometer voltage and current and calculate power in.
- 3) Run the program and set the DC output for the armature to 50% and the output for the field winding to around 50%.
- 4) Complete the table to show speed, output power, dynamometer power and efficiency.

Field winding power supply	Armature winding supply DC2	Speed rpm	Power out	Dyno current	Dyno Voltage	Efficiency
50%	50%					
70%	70%					